

Trastuzumab Emtansine (T-DM1) – Breast Cancer

Drug Summary

Trastuzumab emtansine (T-DM1, trade name **Kadcyla**) is an FDA- and EMA-approved antibody–drug conjugate for **HER2-positive breast cancer** ¹ ². It combines the anti-HER2 antibody trastuzumab with the microtubule inhibitor DM1 (a maytansinoid) via a stable linker. T-DM1 binds HER2 on cancer cells, is internalized, and releases DM1 to induce mitotic arrest and apoptosis ³. Because it requires HER2 overexpression for efficacy, T-DM1 is indicated only in HER2-positive settings (metastatic disease after prior trastuzumab/taxane therapy ¹, or as adjuvant therapy for residual disease after neoadjuvant treatment ⁴). Trastuzumab emtansine has **no activity in HER2-negative subtypes** (e.g. HR+/HER2– luminal cancers or most TNBC/basal tumors) ⁵. Clinical trials (EMILIA, TH3RESA) showed significant PFS/OS benefit in HER2+ metastatic patients relative to lapatinib/capecitabine ⁶, and the KATHERINE trial established its benefit as adjuvant therapy in patients with residual invasive HER2+ disease ⁴. Emerging data (DESTINY-Breast03) show that newer agents (e.g. trastuzumab deruxtecan) can further improve outcomes over T-DM1 ⁷, but T-DM1 remains a key therapy in approved HER2+ indications.

Identifiers & Synonyms

- **ChEMBL:** ChEMBL1743082 (trastuzumab emtansine) (ChEMBL)
- **DrugBank:** DB05773 ²
- **Synonyms/Trade Names:** Kadcyla (brand) ⁵; ado-trastuzumab emtansine; trastuzumab-DM1; trastuzumab-MCC-DM1; T-DM1; TDM1 ⁸; development codes RG-3502, PRO-132365 ⁹.

Mechanism of Action (Breast Cancer Context)

T-DM1 delivers the potent DM1 payload selectively to HER2-overexpressing breast cancer cells. The trastuzumab antibody portion binds the extracellular domain of **HER2 (ERBB2)**, blocking HER2 signaling and eliciting ADCC like trastuzumab ¹⁰ ³. The HER2–T-DM1 complex is internalized via receptor-mediated endocytosis ³. In lysosomes, DM1 (mertansine) is released and binds β -tubulin, inhibiting microtubule polymerization, arresting cells in mitosis and triggering apoptosis ³ ¹¹. Thus, T-DM1 has a dual action: (1) **HER2 blockade**, which suppresses downstream RAS–MAPK and PI3K–AKT–mTOR signaling pathways ¹⁰ (leading to reduced proliferation), and (2) **microtubule cytotoxicity** from DM1, causing cell-cycle arrest and cell death ³ ¹¹. The Fc region of trastuzumab also mediates ADCC (NK-cell cytotoxicity) ³ ¹⁰. Importantly, T-DM1 activity depends on high HER2 expression; tumors with low or heterogeneous HER2 expression show poor T-DM1 efficacy ¹² ¹³. Its effectiveness is largely independent of hormone receptor or BRCA status (HRD); HER2 positivity is the key biomarker for sensitivity.

Primary Targets (Human)

- **ERBB2 (HER2, HGNC:3774)** – extracellular receptor tyrosine kinase bound by the trastuzumab portion of T-DM1 ³ . HER2 blockade underlies the drug's selectivity for HER2+ tumors and initiates its anti-proliferative signal cascade ¹⁰ ³ .
- **TUBB (beta-tubulin, HGNC:12399)** – targeted by the DM1 payload. After internalization, DM1 binds tubulin in the mitotic spindle, inhibiting microtubule function and inducing mitotic arrest and apoptosis ³ .

Annotation: ERBB2 targeting by trastuzumab has sub-nanomolar affinity ¹⁰ . DM1's affinity for tubulin is high (similar to other maytansinoids) and is the basis for the cytotoxic effect ³ .

Pathways (Overview)

Key biological pathways impacted by T-DM1 include:

- **HER2 (ERBB2) Signaling** – Reactome *Signaling by ERBB2* (R-HSA-1227990) and KEGG *ErbB signaling* (hsa04012). HER2 blockade by T-DM1 inhibits downstream proliferative signaling (RAS–MAPK and PI3K–AKT–mTOR pathways) ¹⁰ . Upregulation of these pathways (e.g. activating PIK3CA or RAS mutations) could confer resistance.
- **PI3K–AKT–mTOR Pathway** – Hallmark *PI3K/AKT/mTOR signaling*. Normally downstream of HER2, this survival pathway is suppressed by HER2 inhibition ¹⁰ . Constitutive activation of PI3K–AKT (e.g. PIK3CA mutations) is a known resistance mechanism in HER2+ BC and may reduce T-DM1 sensitivity.
- **RAS–MAPK Pathway** – Hallmark *KRAS signaling*. T-DM1 indirectly downregulates RAS–MAPK signaling via HER2 blockade ¹⁰ . If RAS/MAPK is aberrantly upregulated (e.g. EGFR co-signaling or KRAS activation), cells may bypass HER2 inhibition, leading to resistance.
- **G2/M Cell-Cycle Checkpoint** – Hallmark *G2M checkpoint* and *E2F targets*. DM1's action causes metaphase arrest, engaging the G2/M checkpoint. High expression of cyclin B1 (CCNB1) and other mitotic regulators makes cells more susceptible to DM1; conversely, loss of G2/M drivers (e.g. CCNB1 downregulation) induces T-DM1 resistance ¹⁴ .
- **Apoptosis Pathway** – Hallmark *Apoptosis*. Microtubule disruption and HER2 inhibition trigger intrinsic apoptosis. Key apoptotic effectors (caspases, BCL2 family) determine whether cells undergo T-DM1-induced death; upregulation of anti-apoptotic signals (e.g. BCL2/BCL-XL) could blunt response.
- **Endocytosis/Lysosomal Processing** – Reactome *Endocytosis* and *Lysosome* pathways. T-DM1 requires efficient endocytic uptake and lysosomal proteolysis to release DM1 ³ . Factors affecting vesicle trafficking or lysosomal function (e.g. SLC46A3 transporter) modulate T-DM1 activity ¹⁵ .
- **ABC Drug Transporters** – KEGG *ABC transporters* (hsa02010). T-DM1's DM1 payload can be effluxed by ATP-binding cassette pumps (notably MDR1/ABCB1) ¹⁶ . Upregulation of these transporters in tumor cells leads to rapid drug efflux and resistance, whereas inhibition of MDR1 reverses resistance ¹⁶ .
- **Immune Effector Pathways** – Hallmark *Interferon-γ response* and related immune signatures. T-DM1's Fc region can recruit NK cells and macrophages for ADCC/ADCP ³ ¹⁰ . A pre-existing interferon-γ-rich, TIL-infiltrated tumor microenvironment appears to enhance ADC efficacy. In contrast, high expression of immune checkpoints (PD-1/PD-L1) or immunosuppressive factors (Tregs, CD73) is associated with poorer T-DM1 response, whereas combining T-DM1 with checkpoint inhibitors can restore sensitivity ¹⁷ ¹⁸ .

Upregulated Pathways

Pathways whose activity is **upregulated** in the T-DM1 context (typically in resistant cells) include:

- **ABC Transporter (Drug Efflux) Pathway:** Upregulation of ATP-binding cassette transporters (e.g. MDR1/ABCB1, MRP1) is reported in T-DM1-resistant cells ¹⁶. Enhanced efflux of DM1 reduces intracellular drug concentration, leading to T-DM1 **resistance** ¹⁶.
- **Mitotic Kinases (PLK1) Pathway:** Polo-like kinase 1 (PLK1) is frequently elevated in T-DM1-resistant models ¹⁹. High PLK1 activity promotes mitotic progression independent of HER2 inhibition and dampens T-DM1 cytotoxicity; PLK1 inhibition can resensitize resistant cells. Thus, upregulation of the PLK1/cell-cycle pathway confers **resistance** ¹⁹.
- **Endocytic Recycling:** Some resistant cells exhibit increased recycling of HER2-T-DM1 complexes back to the membrane (rather than lysosomal degradation) ²⁰. Increased HER2 recycling reduces DM1 release, effectively bypassing T-DM1 action, so elevated endosome recycling is a resistance mechanism ²⁰.

Downregulated Pathways

Pathways **downregulated** (or suppressed) in T-DM1 resistance include:

- **HER2 Signaling:** Loss or reduction of HER2 expression/activity is a well-documented resistance mechanism ¹². Downregulation of ERBB2 (through gene heterogeneity or mutation) means fewer targets for T-DM1 binding, thus conferring **resistance** ¹².
- **Cell-Cycle Regulators:** T-DM1 relies on active cell cycling to exert its effect. Knockdown of key mitotic drivers (e.g. CCNB1/cyclin B1 or CDC20) leads to cell-cycle arrest in G2 and prevents DM1 from inducing apoptosis, causing **resistance** ¹⁴. Conversely, maintenance of these pathways is required for sensitivity.
- **Endocytic/Lysosomal Proteins:** Reduction in proteins mediating endocytosis or lysosomal trafficking (e.g. caveolin-1, endophilin A2/SH3GL1) can impair T-DM1 uptake and payload release. Indeed, downregulation of caveolin-1 or SH3GL1 reduces T-DM1 internalization and cytotoxicity ²¹ ²², linking such downregulation to **resistance**.
- **Lysosomal Transporters (SLC46A3):** SLC46A3 is required for lysosomal export of DM1; its loss (downregulation) in resistant cells prevents DM1 release into the cytosol, causing resistance ¹⁵.

Sensitivity Pathways for Upregulation

Pathways that **enhance T-DM1 sensitivity when upregulated** include:

- **Cyclin B1/CDK1 (G2/M Cell-Cycle) Pathway:** High expression of cyclin B1 (CCNB1) and related regulators drives cells into mitosis, where T-DM1's DM1 payload is most effective. Upregulating CCNB1 or CDC20 promotes mitotic entry and sensitizes cells to T-DM1-induced arrest ¹⁴.
- **Immune Effector Pathways:** Enhanced immune activation sensitizes tumors to T-DM1. For example, increased NK cell/NKT cell and cytotoxic T-cell gene signatures (e.g. IFN γ and perforin pathways) correlate with better T-DM1 response ¹⁷ ¹⁸. Experimentally, addition of checkpoint inhibitors (PD-1/PD-L1 or CTLA-4 blockade) to T-DM1 dramatically increased its efficacy ¹⁸, suggesting that upregulation of immune effector pathways (or downregulation of checkpoints) is a sensitivity mechanism.

- **Caveolin-Mediated Endocytosis:** Elevated caveolin-1 (CAV1) expression enhances T-DM1 uptake. In preclinical models, CAV1 overexpression (or stimulation via metformin) increased T-DM1 internalization and sensitivity, whereas CAV1 knockdown reduced sensitivity ²².
- **Endophilin A2 (SH3GL1):** Upregulation of SH3GL1, an endocytic scaffolding protein, promotes efficient HER2-T-DM1 internalization. Knockdown of SH3GL1 markedly suppressed T-DM1 cytotoxicity ²¹, so high SH3GL1 levels likely correlate with greater sensitivity.
- **SLC46A3 (Lysosomal Efflux):** Higher expression of SLC46A3 facilitates DM1 release from lysosomes; therefore, upregulated SLC46A3 promotes DM1 cytotoxicity and sensitivity ¹⁵.

Sensitivity Pathways for Downregulation

Pathways that **enhance sensitivity when downregulated** include:

- **PLK1/Mitotic Exit Pathways:** Downregulation of PLK1 shifts cells toward prolonged mitotic arrest with DM1, increasing apoptosis. Inhibition or knockdown of PLK1 restored T-DM1 sensitivity in resistant cells ¹⁹. Thus, reducing PLK1 activity (and related mitotic checkpoint controls) sensitizes cells to T-DM1.
- **ABC Efflux Transporters:** Downregulation or inhibition of ABC transporters (especially MDR1/ABCB1) prevents drug efflux. Indeed, MDR1 knockdown or pharmacologic inhibition markedly increased sensitivity to T-DM1 (and related ADCs) ¹⁶.
- **Immune Checkpoints:** Downregulation of PD-1/PD-L1 or other immunosuppressive signals (and concomitant increase in immune activation) appears to sensitize tumors. For example, PD-L1 blockade in T-DM1-treated patients showed a trend toward improved PFS ¹⁸, indicating that reducing checkpoint signaling enhances T-DM1 efficacy.

Resistance Pathways for Upregulation

Pathways that **contribute to resistance when upregulated** include:

- **ABC Efflux Transporters:** As above, upregulation of MDR1/P-glycoprotein and related ABC pumps actively exports DM1 out of cells ¹⁶, causing robust **resistance**. Inhibition of these transporters reverses resistance.
- **PLK1/Cyclin-Dependent Kinases:** Overexpression of PLK1 (and possibly Aurora kinases or other mitotic kinases) accelerates mitotic progression and counters T-DM1-induced arrest ¹⁹. High PLK1 is therefore a resistance mechanism; PLK1 inhibitors restore sensitivity.
- **EGFR/HER Family Bypass:** Though less well established, compensatory signaling via EGFR or HER3 could lessen dependency on HER2. (Clinical data on EGFR in T-DM1 resistance are mixed ¹⁰.) However, any upregulation of parallel RTKs or downstream survival kinases would be expected to drive resistance.
- **Immune Checkpoints and Suppression:** Elevated PD-1/PD-L1, CTLA-4, Tregs or CD73 suppress anti-tumor immunity and blunt T-DM1's Fc-mediated effects ¹⁷. High PD-L1 levels have been associated with poorer outcomes on HER2-targeted therapy; thus, upregulation of immunosuppressive pathways fosters resistance.

Resistance Pathways for Downregulation

Pathways that **cause resistance when downregulated** include:

- **HER2/ERBB2 Signaling:** Reduction or loss of HER2 expression abrogates T-DM1 binding, causing resistance ¹². Tumors with HER2 heterogeneity or low HER2 expression fail to respond to T-DM1 ¹² ¹³.
- **Cell-Cycle Entry (Cyclin B1):** As noted, downregulation of CCNB1 or CDC20 prevents cells from entering mitosis where DM1 acts, leading to resistance ¹⁴.
- **Endocytosis/Lysosomal Functions:** Knockdown of key endocytic proteins (CAV1, SH3GL1) or lysosomal enzymes prevents T-DM1 trafficking to its site of action ²¹ ²². For instance, reduced caveolin-1 or endophilin levels markedly suppress T-DM1 uptake and cytotoxicity.
- **Lysosomal Transport (SLC46A3):** Loss of SLC46A3 expression locks DM1 inside lysosomes, preventing cell kill ¹⁵.

Breast Cancer Subtype–Stratified Evidence

Trastuzumab emtansine efficacy is strictly tied to **HER2 status**. It is effective in all HER2-positive subtypes (including HER2-enriched or Luminal B/HER2+ cancers) and is indifferent to hormone receptor or BRCA/HRD status per se. In contrast, **HR-positive/HER2-negative (Luminal A/B)** and **triple-negative/basal-like** tumors express little or no HER2 and are not targets of T-DM1 ⁵. No trial evidence supports T-DM1 in these HER2-negative subtypes. Notably, intratumoral heterogeneity of HER2 (e.g. mixed HER2+/HER2– regions) predicts T-DM1 failure ¹² ¹³. Germline BRCA1/2–mutated tumors are often triple-negative, and the absence of HER2 in these cases means T-DM1 is inactive; there is no known interaction between HRD status and T-DM1 sensitivity.

Contraindications and Safety

Contraindications: There are no absolute contraindications listed in the label, aside from hypersensitivity to T-DM1 components. However, T-DM1 carries *black-box warnings* for serious toxicities ²³ ²⁴. It can cause severe hepatotoxicity (life-threatening liver failure) and significant cardiac toxicity (reduced LVEF) ²³. Embryo-fetal risk is also high; it is contraindicated in pregnancy (Category D/X) ²³. Clinicians must not substitute T-DM1 for trastuzumab and must monitor LFTs and cardiac function closely ²³.

Safety: The most common adverse events ($\geq 25\%$) in T-DM1–treated patients include fatigue, nausea, thrombocytopenia, transaminase elevations, anemia, musculoskeletal pain, hemorrhage, headache, and peripheral neuropathy ²⁵. These mirror its dual action (trastuzumab-like cardiac events; DM1-related myelosuppression). Toxicities are generally manageable but can impact quality of life (e.g. neuropathy, thrombocytopenia). T-DM1 has a more favorable GI toxicity profile than oral HER2 TKIs (as seen in EMILIA) ²⁶. Pulmonary toxicity (pneumonitis) is rare but reported. Trastuzumab emtansine should be avoided in patients with active liver disease or significant left ventricular dysfunction, in line with its warning profile.

Trial and Guideline Context (FDA/EMA/NCCN/ESMO/ASCO)

- **FDA/EMA Approvals:** T-DM1 was first approved in 2013 for HER2+ metastatic breast cancer after prior trastuzumab/taxane therapy ¹ ²⁷. In May 2019, the FDA granted approval for adjuvant use

in patients with residual invasive HER2+ disease post-neoadjuvant therapy, based on the KATHERINE trial (3-year invasive DFS 88.3% vs 77.0%) ⁴ . EMA followed similarly. T-DM1 is now standard for these indications.

- **Clinical Trials:** Key Phase III trials include EMILIA (T-DM1 vs lapatinib-capecitabine in 2nd-line MBC ²⁶), TH3RESA (T-DM1 vs physician's choice in heavily pretreated MBC ²⁸), MARIANNE (T-DM1 ± pertuzumab vs trastuzumab + taxane, showing similar efficacy but better tolerance) ¹⁰ , and KRISTINE (T-DM1+pertuzumab vs standard chemo+HER2 therapy in neoadjuvant, showing inferior pCR with T-DM1 combination) ²⁹ . The KATHERINE trial (adjuvant residual disease) showed a dramatic 50% risk reduction in invasive disease/death with T-DM1 over trastuzumab ⁴ . More recent trials (e.g. DESTINY-Breast03) have established trastuzumab deruxtecan as superior to T-DM1 in second-line metastatic HER2+ disease ⁷ .
- **Guidelines:** Current breast cancer guidelines incorporate T-DM1 in its approved niches. For example, NICE recommends (TA632) T-DM1 as adjuvant therapy for HER2+ early breast cancer with residual invasive disease after neoadjuvant therapy ³⁰ . NCCN guidelines (2024) list adjuvant T-DM1 (14 cycles) for residual disease (Category 1 evidence) and as a preferred agent in the metastatic second-line setting (although newer ADCs like T-DXd are also recommended). ESMO and ASCO similarly endorse these indications. All guidelines stress that HER2 positivity is required; there is no guideline role for T-DM1 in HER2-negative tumors. Recent updates have started to elevate trastuzumab deruxtecan over T-DM1 for the metastatic setting ⁷ , but T-DM1 remains a comparator and is still recommended where T-DXd is not used.

Additional Notes

- **Subtype specificity:** T-DM1's efficacy varies within HER2-positive subgroups. Data suggest somewhat better responses in "HER2-enriched" (non-Luminal) tumors, whereas ER+/HER2+ cancers may have marginally lower pCR rates, possibly due to concurrent ER signaling. However, no formal approval restricts use by HR status. Mutational status (e.g. PIK3CA, TP53) may modulate T-DM1 benefit but have not changed guideline indications.
- **Resistance circumvention:** Combinations to overcome resistance are under study. For instance, T-DM1 plus pertuzumab did not improve efficacy in early studies (MARIANNE, KRISTINE) ¹⁰ ²⁹ . Checkpoint inhibitors (e.g. pembrolizumab, atezolizumab) are being tested with T-DM1 to augment immune response (NCT03032107) ¹⁸ . Switching to alternate ADCs (such as T-DXd) or continuing HER2 blockade with other HER2-targeted therapies are strategies after T-DM1 progression.
- **Pharmacology:** T-DM1 is given IV every 3 weeks at 3.6 mg/kg. It is not interchangeable with trastuzumab (different dosing/regimens) ²³ . Its clearance is slower than trastuzumab due to the DM1 payload; DM1 metabolism partly involves CYP3A enzymes, so strong CYP3A inhibitors/inducers can affect exposure (per label).
- **Biomarkers:** Apart from HER2, no predictive biomarkers for T-DM1 sensitivity/resistance have entered clinical use. Markers under investigation include SLC46A3 expression, Fcγ receptor polymorphisms (affecting ADCC), and various gene signatures of endocytosis or immune activation.

Pathway Evidence Table

Pathway (ID/ Name)	Regulation	Drug Effect	Rationale (Mechanism)	Reference(s)
Reactome R-HSA-1227990 (<i>ERBB2 signaling</i>)	Down	Sensitive	T-DM1 binding inhibits HER2-driven RAS/MAPK and PI3K/AKT signaling, halting proliferation ¹⁰ . Loss of this inhibition confers resistance.	¹⁰ ¹²
KEGG:hsa04012 (<i>ErbB signaling</i>)	Down	Sensitive	Similar to above: HER2 blockade suppresses downstream pro-survival signals.	¹⁰
Hallmark <i>PI3K/AKT/mTOR signaling</i>	Down	Sensitive	Indirectly inhibited by HER2 targeting. Constitutive PI3K/AKT activation blunts T-DM1 effect, causing resistance.	¹⁰
Hallmark <i>KRAS signaling up</i>	Down	Sensitive	HER2 blockade reduces RAS/MAPK signaling ¹⁰ . KRAS/RAS hyperactivation drives resistance by bypassing HER2.	¹⁰
Hallmark <i>G2M Checkpoint</i>	Up	Sensitive	Upregulation (high CCNB1/CDC20) means active cycling into mitosis; cells arrested by DM1-mediated mitotic blockade ¹⁴ .	¹⁴
Hallmark <i>E2F Targets</i>	Up	Sensitive	E2F activation reflects cell-cycle progression; T-DM1 requires dividing cells. CCNB1 suppression reduces efficacy ¹⁴ .	¹⁴
Hallmark <i>Apoptosis</i>	Up	Sensitive	Activation of apoptotic programs (caspases, etc.) enables cell death after DM1 damage.	³
Hallmark <i>Interferon Gamma Response</i>	Up	Sensitive	Reflects IFN γ -driven immune activation. Enhanced TIL/IFN γ signaling correlates with better ADC response and can prime ADCC ¹⁷ ¹⁸ .	¹⁷ ¹⁸

Pathway (ID/ Name)	Regulation	Drug Effect	Rationale (Mechanism)	Reference(s)
Reactome <i>Endocytosis</i>	Up	Sensitive	Increased receptor-mediated uptake of HER2-T-DM1 complex leads to more DM1 release. Loss of endocytosis components causes resistance.	3 21
Reactome <i>Lysosome</i>	Up	Sensitive	Efficient lysosomal degradation of T-DM1 releases DM1 payload. Impaired lysosomal function (e.g. SLC46A3 loss) causes resistance ¹⁵ .	3 15
Reactome <i>ABC transporters</i> (KEGG 02010)	Up	Resistant	Overexpression of MDR1/P-gp (ABCB1) and related pumps effluxes DM1, reducing intracellular drug ¹⁶ . Upregulation confers resistance.	16
Reactome R-HSA-5663201 (<i>RAS activation</i>)	Up	Resistant	Hyperactivation of RAS/MAPK (e.g. via RTK bypass) overcomes HER2 blockade, promoting survival and resistance.	10
Hallmark <i>DNA Repair</i>	Up	Resistant	(Speculative) Enhanced DNA repair might counter DNA damage from any secondary effects of DM1; not well-established for T-DM1 specifically.	– (no direct evidence)
Hallmark <i>TNFα/NF-κB signaling</i>	Up	Resistant	Activation of NF-κB can promote survival and drug resistance in many settings; may blunt ADC-induced apoptosis.	–
GO: <i>Caveola-mediated endocytosis</i>	Up	Sensitive	Increased caveolin-1 (CAV1) enhances T-DM1 uptake in some models ²² . Downregulation lowers sensitivity.	22
GO: <i>Microtubule cytoskeleton organization</i>	Down	Resistant	Reduced microtubule assembly factors (TUBB etc.) may impair DM1 binding, although evidence is limited.	–

Pathway (ID/ Name)	Regulation	Drug Effect	Rationale (Mechanism)	Reference(s)
Reactome R-HSA-69409 (Ubiquitin-mediated proteolysis)	Up	Resistant	Upregulation of ubiquitination pathways (e.g. altered lysosomal targeting) could degrade HER2-T-DM1 complexes before DM1 release, impairing efficacy.	- (emerging)
Hallmark <i>PI3K/AKT/mTOR signaling</i>	Up	Resistant	(As above) Constitutive activation bypasses HER2 blockade.	10
Hallmark <i>Estrogen Response</i>	-	-	Not directly modulated by T-DM1; influences sensitivity only via effect on HER2+ luminal tumors, not a direct T-DM1 pathway.	-

Each entry above is based on human breast cancer-specific data and mechanistic rationale. “Up/Down” refers to pathway activity change; “Sensitive” means pathway status correlates with drug efficacy, “Resistant” means it correlates with lack of response. Evidence is drawn from cited studies and pathway databases (Reactome, KEGG, MSigDB).

Sources: Mechanistic claims and pathway annotations are supported by peer-reviewed studies and authoritative databases. Key references include the DrugBank trastuzumab emtansine entry ³, a 2019 review of T-DM1 resistance mechanisms ^{10 12 17 21}, and FDA/NCCN/ASCO guidance summaries ^{4 30}. All content is based on human data.

^{1 6 10 11 12 13 14 15 16 17 18 19 20 21 22 26 28 29} Mechanisms of resistance to trastuzumab emtansine (T-DM1) in HER2-positive breast cancer | British Journal of Cancer

https://www.nature.com/articles/s41416-019-0635-y?error=cookies_not_supported&code=c83e8097-a223-4887-90f9-2449e06f9b38

^{2 3 5 8 9 23 24} Trastuzumab emtansine: Uses, Interactions, Mechanism of Action | DrugBank Online

<https://go.drugbank.com/drugs/DB05773>

^{4 25} FDA Approval Summary: Ado-Trastuzumab Emtansine for the Adjuvant Treatment of HER2-positive Early Breast Cancer - PubMed

<https://pubmed.ncbi.nlm.nih.gov/32217612/>

⁷ Trastuzumab deruxtecan versus trastuzumab emtansine in HER2-positive metastatic breast cancer: long-term survival analysis of the DESTINY-Breast03 trial | Nature Medicine

https://www.nature.com/articles/s41591-024-03021-7?error=cookies_not_supported&code=3b7fc151-b354-4288-906b-3cd9c82c4f71

²⁷ KADCYLA (ado-trastuzumab emtansine) Label

https://www.accessdata.fda.gov/drugsatfda_docs/label/2013/125427lbl.pdf

30 1 Recommendations | Trastuzumab emtansine for adjuvant treatment of HER2-positive early breast cancer | Guidance | NICE

<https://www.nice.org.uk/guidance/ta632/chapter/1-Recommendations>